Anjali Aggarwal
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Education

Class X (CBSE)	Ramjas School	*2019, 86.89%*
Class XII (CBSE)	St. Anthony School	2021, 87%
B.Tech (IT)	Dr. Akhilesh Das Gupta Institute of Professional Studies, GGSIPU	2021–2025,91%
Diploma in French	University of Delhi, College of Arts & Sciences (German and Science College)	Aug 2025 – Mar 2026

Professional Summary

AI/ML Enthusiast and Full-Stack Developer with hands-on experience in building **scalable machine learning models and web applications**. Proficient in **Python, PyTorch, TensorFlow, and SQL**, with strong background in **data preprocessing, model evaluation, and deployment pipelines**. Experienced in both AI/ML projects (classification, NLP, explainability, MLOps) and full-stack web development (React.js, Node.js, REST APIs, Docker). Adept at transforming large, complex datasets into **actionable insights and intelligent solutions**. Currently developing an **Explainable AI system for medical X-ray classification**, showcasing ability to deliver end-to-end, responsible AI pipelines.

Core Skills

- **Machine Learning & AI:** PyTorch, TensorFlow, Transformers, CNNs, Transfer Learning, NLP (BERT, spaCy, NLTK), Explainable AI (Grad-CAM), LLM Fine-tuning
- **Programming & Data:** Python, SQL, Data Preprocessing & EDA
- **Analytics & Visualization:** Matplotlib, Seaborn, Tableau/Power BI, Model Evaluation (Confusion Matrix, ROC, F1)
- **MLOps & Deployment:** FastAPI, Docker, React.js, Git
- **Web Development:** HTML, CSS, JavaScript, Node.js, MongoDB
- **Vector Databases & RAG Pipelines:**FAISS, Pinecone, Weaviate, LangChain, Sentence-Transformers, RAG Optimization, Contextual Search

Experience

KVon Technologies – Machine Learning Trainee

May 2025 – Present

- ✓ Undergoing intensive ML training with real-world projects, focusing on **supervised and unsupervised learning**, model evaluation, and problem-solving using **Python (NumPy, Pandas, Scikit-learn)**.
- ✓ Applied advanced **statistical modeling** and **data analytics** to solve business challenges.
- ✓ Designed data pipelines for automation and actionable insights

Zobox Protect Limited – Web Developer

Dec 2024 – Apr 2025

- ✓ Contributed to full-stack development of responsive **web solutions using MERN stack and Next.js**, with real-time updates and secure authentication.
- ✓ Improved scalability and performance of existing projects by optimizing front-end and back-end modules.
- ✓ Created **interactive dashboards** for data visualization and reporting, enhancing decision-making.

Code Alpha – Web Development Intern

Jun 2024 – Sep 2024

- ✓ Built a **music web browser** with playlist management and search, integrating **React.js, JavaScript, and REST APIs**.
- ✓ Enhanced UI/UX with responsive design and data-driven features.
- ✓ Increased session time by **35%** through user feedback-driven enhancements.

Projects

Multimodal Image & Voice QA System (FastAPI + YOLO + BLIP + Whisper)

- ✓ Built a local multimodal assistant that can **analyze images, detect people/objects, generate captions, and answer questions via text or voice**.
- ✓ Integrated **YOLOv8, BLIP, and Whisper** for detection, captioning, and speech-to-text—all running fully offline.
- ✓ Developed a FastAPI backend with a lightweight HTML/JS frontend supporting **image upload + webcam capture + mic recording**.
- ✓ Implemented automatic **image annotation** using OpenCV and stored metadata for follow-up queries.
- ✓ Delivered a fast, privacy-preserving system that enables interactive visual understanding without cloud dependencies.

Medical X-Ray Classification with Explainable AI

- ✓ Developed a **PyTorch (DenseNet-121)** model for **multi-class chest X-ray classification** (COVID-19, Pneumonia, Lung Opacity, Normal) using the **COVID-19 Radiography Dataset (~18,000 images)**.
- ✓ Achieved **~90% accuracy** with robust preprocessing (resizing, normalization, augmentation), optimized training pipeline (AdamW, LR scheduling, early stopping).
- ✓ Integrated **Grad-CAM heatmaps** for model explainability, improving transparency for clinical validation.
- ✓ Deployed with **Flask/FastAPI, React.js, and Docker** for scalable cloud inference.
- ✓ Embedded **RAG pipeline** for intelligent report summarization using **vector database (FAISS)** and **transformer-based text generation**, enhancing interpretability and automation.

Fake News Detector (NLP + Machine Learning)

- ✓ Built a **Fake News Detection System** using NLP and ML algorithms (**PassiveAggressiveClassifier, TF-IDF**).
- ✓ Preprocessed text data with **NLTK**, engineered features, and achieved **96% accuracy**.
- ✓ Deployed the model via **Streamlit** for real-time predictions.

JarvisAI – Desktop Bot

- ✓ Engineered a **voice-controlled desktop assistant** in Python to automate daily tasks (web search, reminders, app launch).
- ✓ Integrated speech recognition and natural language processing for intelligent interactions.

Certifications

- ➢ Certificate in Data Science – IBM
- ➢ Specialization in AI/ML

Languages

Hindi (Native), English (Proficient), French (Intermediate, pursuing Diploma)